

SECTION C

10 (short)

(a) Describe what is meant by:

- (i) magnetic flux, and
- (ii) magnetic flux density B .

How are these concepts relevant to flux in the electromagnet sketched in Fig. 6? [5]

(b) Calculate the current required to produce $B = 0.2 \text{ T}$ in the air gap if $l_g = 2 \text{ mm}$, $l_c = 40 \text{ mm}$, $N = 200$ turns and the electromagnet is made from a linear magnetic material with relative permeability 250. State any approximations made. [5]

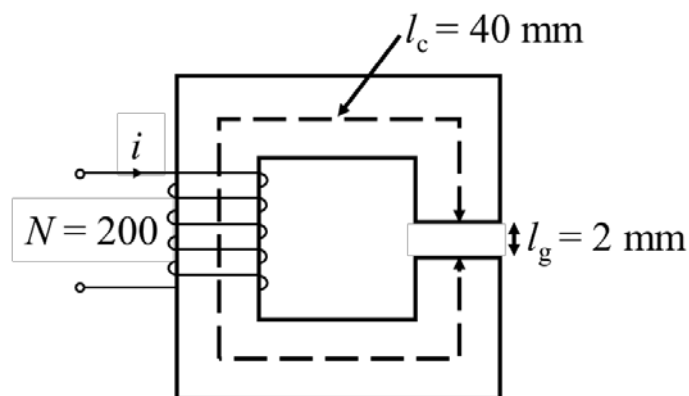


Fig. 6